1 Target Architecture — Grounding AI in Human Workshop Reality

Built for a real African machine shop: runs 100% offline, guarantees physical safety, and leaves the machinist in command. NEW = 2-week sprint commitment | HUMAN = operator validation gate | ESTOP = 0 ms hardware cut

Machinist Interface & Voice Assistant

Flutter App — Workshop PC (Windows) & Floor Tablet (Android)

- **Shop Voice & Chat** — Natural French, Bambara, Swahili, etc.
- **Confirmation Card** HUMAN — Mandatory tap before motor/spindle move
- **3D Toolpath Preview** HUMAN — Visual simulation before cutting metal
- **Local / Cloud Privacy Pill** — Operator sees exactly where data stays
- **AI Health Monitor** NEW — Real-time warning if AI slows or hangs

▲ Spoken / typed operator questions • Verified recommendations & cards ▼

Safe Gateway & Control Pipeline

Deterministic application core — guards every single turn

1 · E-STOP Bypass (0 ms software cut) › 2 · Lexicon Filter (Safe read-only routing) › 3 · Grounded Context (Sensors + manuals) › 4 · Tool Router (Restricted capabilities) › 5 · Output Guards (Blocks raw G-code) › 6 · Kinematics (Travel & speed checks)

- **30 s Timeout & Circuit Breaker** NEW — Never leaves the operator waiting on shop floor
- **Enforced Machine State Read** NEW — Must poll hardware encoders before answering position
- **Auto Doc Pre-fetch** NEW — Grounds troubleshooting in verified manuals
- **Model Digest Verification** NEW — Blocks boot if weights or prompt changed

▲ Local private HTTP (:11434) • Optional 4G cloud fallback strictly on demand ▼

Local Workshop Inference (Air-Gapped LAN)

Shop PC — Ollama :11434 • RTX 3070 8 GB • $0 Cloud Cost

- **forgeron-qwen (Qwen 2.5 7B)** — Custom Modelfile • 16k context • Pinned hash NEW
- **Visual Inspector VLM** — Checks tool wear, swarf, clamp setup
- **Local Dialect Engine** — Adapts French to workshop vernacular
- **Python 3D CAD Engine** — OpenCASCADE calculates geometry — never LLM

Cloud Fallback (Optional)

- **Gemini 2.5 Flash Lite** — Used only if operator enables 4G; pinned model version NEW

▲ Encoders, machine coordinates, tools catalog, and French documentation ▼

Data Ingestion & Grounding Sources

Sanitized & validated locally before touching neural model

- **Live Machine Coordinates** — Duet3D/ESP32 polled via WebSocket
- **Workshop Manuals (RAG)** — Markdown/PDF docs versioned in git
- **Audio & Accents Normalizer** — Cleans shop background spindle noise
- **STEP / STL CAD Files** — Strict path & integrity verification NEW
- **Locked Shop Lexicon** — Deterministic dictionary of machine terms

Physical Machine Zone: Duet3D / FluidNC → 5-Axis Milling Center

Motors move ONLY after safety validators pass AND machinist explicitly confirms on screen.

HARDWARE E-STOP: 0 ms Instant Power Cut

HUMAN-CENTERED SAFETY

- **Human-in-the-Loop Imperative:** The AI is a digital apprentice. It advises; the human operator decides and confirms. Zero autonomous moves.
- **Physical Protection:** Hard software envelopes prevent moves outside machine stroke (Z-bed floor, B-trunnion limits).
- **Instant Emergency Stop:** Red physical mushroom button is hardwired to power relays, completely independent of all AI software.

GRACEFUL DEGRADATION LADDER

1. **Tier 1 · Local AI Assistant:** Rich conversational guidance and visual diagnosis running offline on workshop GPU.
2. **Tier 2 · Deterministic Lexicon:** If AI hangs or model crashes, keyword lookup provides instant answers (< 1 s).
3. **Tier 3 · Manual CNC Screen:** Direct DRO & manual jog controls keep machine producing parts with zero downtime.

OPERATIONAL TELEMETRY

- **Local AiTurnJournal:** Structured JSONL on disk logs latencies, token usage, tool calls, and safety stops. *Zero operator text/audio recorded.*
- **Release Fingerprint Gate:** CI verifies Model digest + prompt hash + tool schemas before any workshop deployment.

2 Failure Register — Protecting the Human & Machine from AI Errors

In a machine shop, AI bugs cause broken carbide, ruined workpieces, or physical danger. Scored with FMEA: RPN = Likelihood × Impact × Detection (1–5 scale).

Failure Mode & Workshop Reality	RPN	How We Catch It Early (Signals & Alerts)	Preventive Engineering	Fast Recovery & Rollback Runbook
A · SILENT DRIFT **The Model Changes Behind Our Back** Someone pulls new Ollama weights, tweaks prompt formatting, or updates tools without benchmarking. Feeds & speeds degrade silently until a $150 cutter snaps. **Audit Proof:** Release gate was skipped; forgeron-qwen tag is mutable; 500 uncommitted AI code lines in working tree; no automated CI pipeline.	**80** L4 × I4 × D5	• **P1 Alert:** Startup model digest ≠ release manifest → local AI blocked, falls back to lexicon. • **P2 Alert:** Runtime prompt hash drift → warning banner shown to operator. • **CI Gate:** Release test skipped or failed → build automatically breaks. • **Benchmark:** Overall score drops > 5% or any safety regression blocks promotion.	Pin exact model hash and Modelfile in git; enforce automated CI on every commit; require full 85-scenario benchmark before shop promotion.	• Run `tool/ai_rollback.ps1 ai-vN` to restore verified prompt, tools, and model weights. • Re-verify digest and run release test. • Smoke test safety scenarios (~10 min). • **Target: ≤ 15 min** • *Owner: AI Lead*
B · UNGROUNDED GUESS **Hallucinating Machine State** The machinist asks: *"Can I safely clear the B-axis clamp?"*. The AI answers from conversation memory instead of reading live encoders, risking a catastrophic crash into the trunnion table. **Benchmark Proof:** Freshness suite failed 60%; Documentation suite failed 62.5%. Model guessed Z-height from stale history.	**64** L4 × I4 × D4	• **P2 Alert:** Axis position or status answer emitted without fresh `get_machine_state` call > 2% daily. • **P2 Alert:** Troubleshooting query answered without calling `search_docs` > 10% daily. • **Benchmark Gate:** Freshness score < 80% or Documentation score < 70% rejects release candidate.	Deterministic pre-fetching in application core: force an encoder read before generating position answers; auto-fetch maintenance docs for "how-to" questions.	• Operator switches UI to **Lexicon-only mode** in settings: status answers become 100% deterministic (**< 1 min**). • Roll back to prior release if drift persists. • Add failing phrasing to golden bench suite. • *Owner: AI Lead + Operator reports*
C · DANGEROUS ADVICE **Drafting Raw G-Code or Out-of-Stroke Moves** The operator asks how to face a block; the AI emits 15 lines of raw G-code. A single negative Z typo would plunge the spindle at 24,000 RPM directly into the machine bed. **Benchmark Proof:** No-G-Code rule failed 73%; model invented fake file paths (`/path/to/part.step`) and obeyed social engineering.	**60** L3 × I5 × D4	• **P1 Alert:** Any kinematic envelope limit violation followed by operator manual override. • **P2 Alert:** Any raw G-code block detected by output parser in assistant response. • **Benchmark Gate:** Zero tolerance: any forbidden motion command in safety suite immediately rejects release.	Completely eliminate G-code drafting instructions from prompt; reject non-existent CAD paths; enforce mandatory confirmation modal with toolpath visualizer on screen.	• **Machinist hits physical red E-STOP** (0 ms instant power cut). • Toggle AI off → switch to manual CNC interface (**< 1 min**). • Incident logged in journal; blameless review conducted within 24 hours. • *Owner: Machinist (Floor) + AI Lead*

Human Safety Watch List: Missed French/Swahili E-STOP voice commands (Severity 5: kept on active watch list regardless of statistical score) • Heavy inference latency spikes (> 15 s locks operator attention on screen) • Safety warnings translated into local vernacular without native speaker sign-off • Unprotected Ollama LAN port exposed to shop Wi-Fi.

3 Tooling & 2-Week Sprint — Building What 2 People Can Truly Maintain

Lean, pragmatic MLOps for a real workshop team (1 AI Developer + 1 Master Machinist). Zero enterprise bloat, zero server maintenance, 100% focus on physical safety.

Pragmatic Toolstack (Zero Bloat)

2-Person Team Reality

Function	Chosen Tool	Why It Fits Our Reality
Code & CI	GitHub + Actions	Free, automated, runs tests on push without maintaining any servers
AI Versioning	`ai_release.json` + git tags	Locks code, prompts, tool schemas, and model digest into one release ID
Evaluation Data	Golden scenarios in git	Fingerprinted test suite; test reports committed directly to repository
Test Runner	`flutter test` + pytest	Runs 69 test files + CAM geometry tests in under 2 minutes
Local Inference	Ollama on Workshop PC	Runs 100% offline, $0 API cost, leverages existing 8 GB GPU
Telemetry	AiTurnJournal + UI panel	Offline JSONL logging; monitors latency & safety stops without storing chat text
Model Bench	Forgeron-Bench	Evaluates 85 real shop scenarios locally with 95% Wilson confidence intervals
Fast Rollback	1-line script + UI toggle	Reverts model release in ≤ 15 min; toggles AI off to manual mode in < 1 min

Deliberately omitted: Kubernetes, hosted MLflow clusters, expensive cloud APM. A 2-person team cannot afford full-time DevOps maintenance; tools must survive shop power cuts.

2-Week Human-Centered Backlog (12 → 25 Nov 2026)

Priority = Risk Reduction ÷ Effort

#	Task Description	Risk	Eff.	Owner	Due	Definition of Done (Testable)
1	Promote verified baseline release; make skipped tests break CI	A	S	AI Lead	12 Nov	`ai_release_test` passes (not skipped); report committed to git
2	Benchmark or revert 500 uncommitted AI code lines	A	S	AI Lead	13 Nov	Full benchmark run: zero safety regressions; clean git status
3	Setup GitHub Actions: automated analysis and unit tests on push	A	S	AI Lead	14 Nov	Green check on main; deliberately broken test turns build red
4	Pin model digest and Gemini version; enforce startup check	A	M	AI Lead	17 Nov	Unit test: launching with modified model hash forces lexicon fallback
5	Implement `ai_rollback.ps1` and run live rehearsal drill	A	M	AI Lead	18 Nov	Recorded drill proving full release rollback in ≤ 15 minutes
6	Enforce deterministic machine state read before position answers	B	M	AI Lead	19 Nov	Forgeron-Bench Freshness score jumps from 40% to ≥ 80%
7	Enforce automatic documentation pre-fetch for diagnostic queries	B	M	AI Lead	20 Nov	Forgeron-Bench Documentation score jumps from 37.5% to ≥ 70%
8	Remove G-code drafting from prompt; validate input file paths	C	M	AI Lead	21 Nov	No-G-Code score reaches ≥ 90%; fake paths immediately rejected
9	Build in-app AI Health Dashboard from local journal logs	B C	M	AI Lead	24 Nov	Widget test passes; shows 24h error rate, p95 latency, safety stops
10	Native review of 20 French safety scenarios & sign incident runbook	all	S	Technician	25 Nov	20 scenarios stamped “reviewed” by named speaker; runbook signed

Sprint Exit Gate: Green CI build, locked release hash, verified 15-minute rollback drill, and benchmark accuracy ≥ 80% with zero safety regressions. Signed off jointly by AI Lead and Master Machinist.